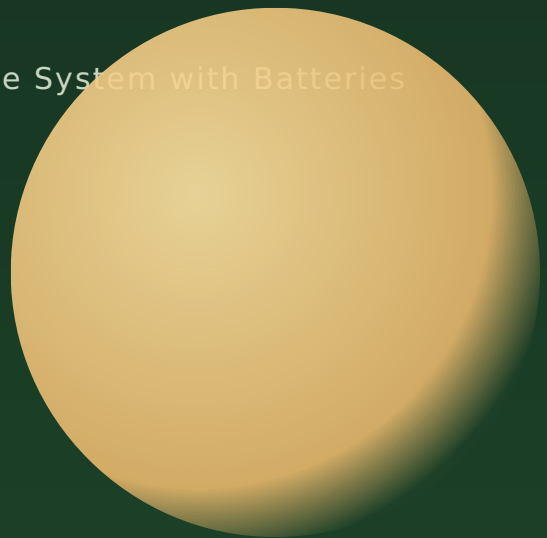


Off-Grid Standalone PV System Report

Design & Performance Simulation — Standalone System with Batteries



480 Wp

System Power

75%

Performance Ratio

93%

Solar Fraction

Project

Location

Simulation Software

Simulation Date

Author

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Dakar / Taklou — Senegal

PVsyst V7.4.8

05/05/2026

01 Project Overview

This report summarizes the simulation results of a standalone (off-grid) photovoltaic system, performed using PVsyst software version 7.4.8, designed to supply a residential household with electricity using solar panels and batteries only, without relying on the public electricity grid. Dakar/Taklou in Senegal was selected as the reference site for the simulation, based on solar irradiation data available from the MeteoNorm database.

02 System Specifications

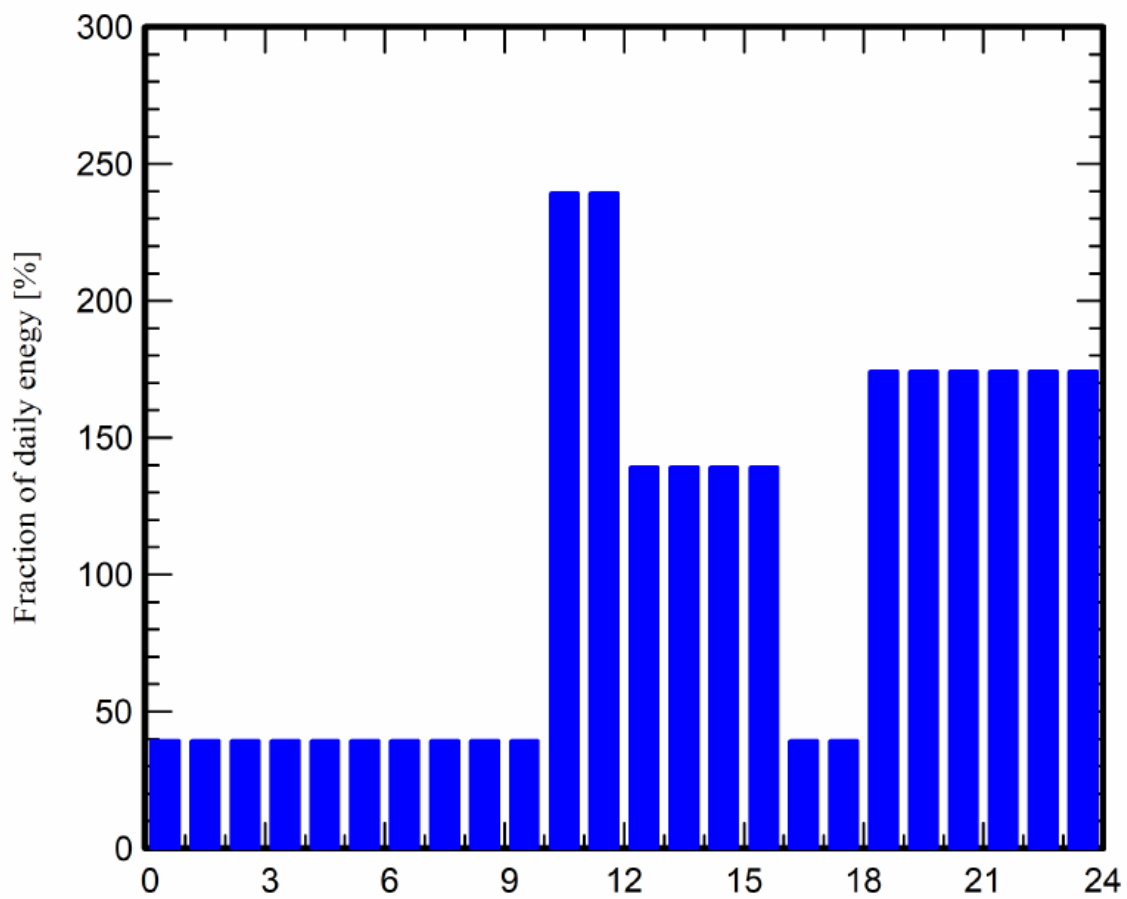
Number of PV Modules	8 modules
Power per Module	60 Wp
Total System Power	480 Wp
Module Configuration	4 in series × 2 in parallel
Tilt / Azimuth	15° / 0°
Controller	MPPT — 97% efficiency

Battery Technology	Lead-acid, Gel
Number of Batteries	4 units (2 series × 2 parallel)
System Voltage	24 V
Total Capacity	320 Ah
Stored Energy	6.1 kWh
Average Daily Consumption	2.3 kWh/day

03 User's Energy Needs

The load was designed assuming an average daily household consumption of 2.3 kWh/day, with seasonal variation based on lighting hours and usage patterns across the four seasons, as shown in the table below:

Item	Summer	Autumn	Winter	Spring
LED Lighting	240	300	360	300
TV / PC / Mobile	225	300	450	300
Domestic Appliances	600	800	800	800
Fridge / Deep-freeze	799	799	799	799
Stand-by Consumers	144	144	144	144
Total Daily Energy (Wh)	2009	2343	2553	2343



Hourly distribution of energy consumption throughout the day (% of daily consumption)

04 Main Simulation Results

784

kWh/year
Useful Energy to User

74.9%

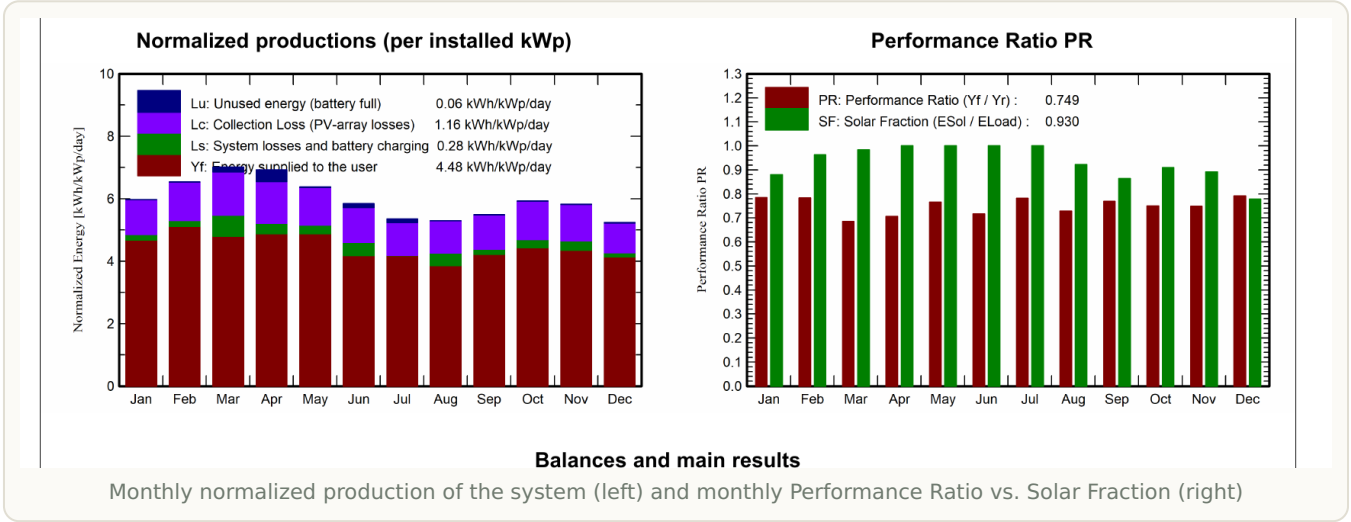
Performance Ratio
(PR)

93.0%

Solar Fraction
(SF)

59.4

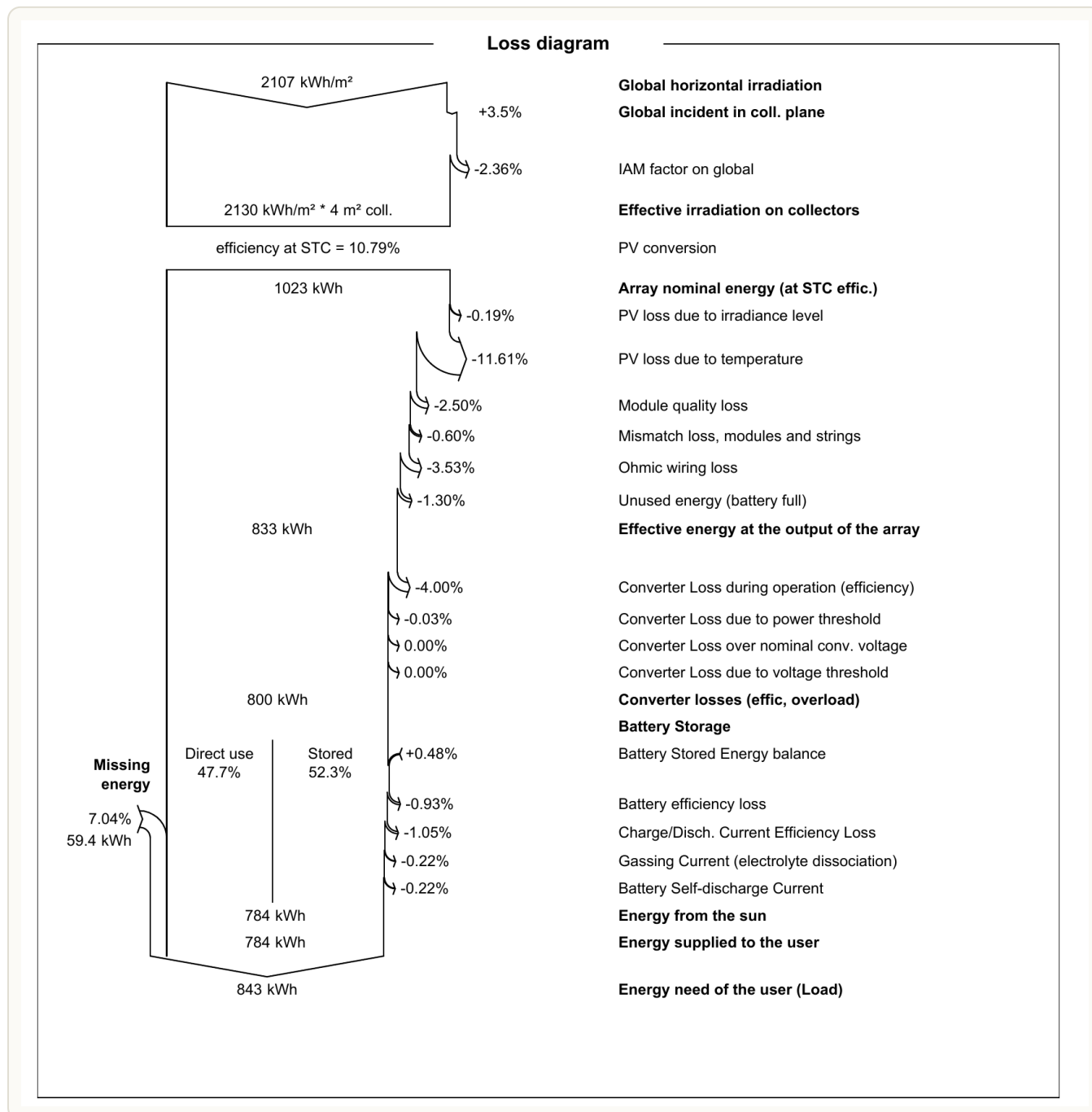
kWh/year
Missing Energy



Month	GlobHor	GlobEff	E_Avail	E_Miss	E_User	SolFrac
January	159.5	181.0	69.5	9.5	69.7	0.88
February	164.7	179.1	68.4	2.7	68.8	0.96
March	207.0	212.8	80.5	1.2	71.5	0.98
April	209.7	202.9	77.3	0.0	70.3	1.00
May	209.2	192.6	73.8	0.0	72.6	1.00
June	188.0	170.5	65.5	0.0	60.3	1.00
July	176.4	161.3	61.5	0.0	62.3	1.00
August	168.7	159.9	60.8	4.8	57.5	0.92
September	161.7	160.7	60.5	9.6	60.7	0.86
October	170.8	179.7	67.1	6.6	66.1	0.91
November	153.1	170.8	64.4	7.6	62.7	0.89
December	137.9	158.3	61.2	17.6	61.6	0.78
Annual Total	2106.7	2129.6	810.5	59.4	784.1	0.93

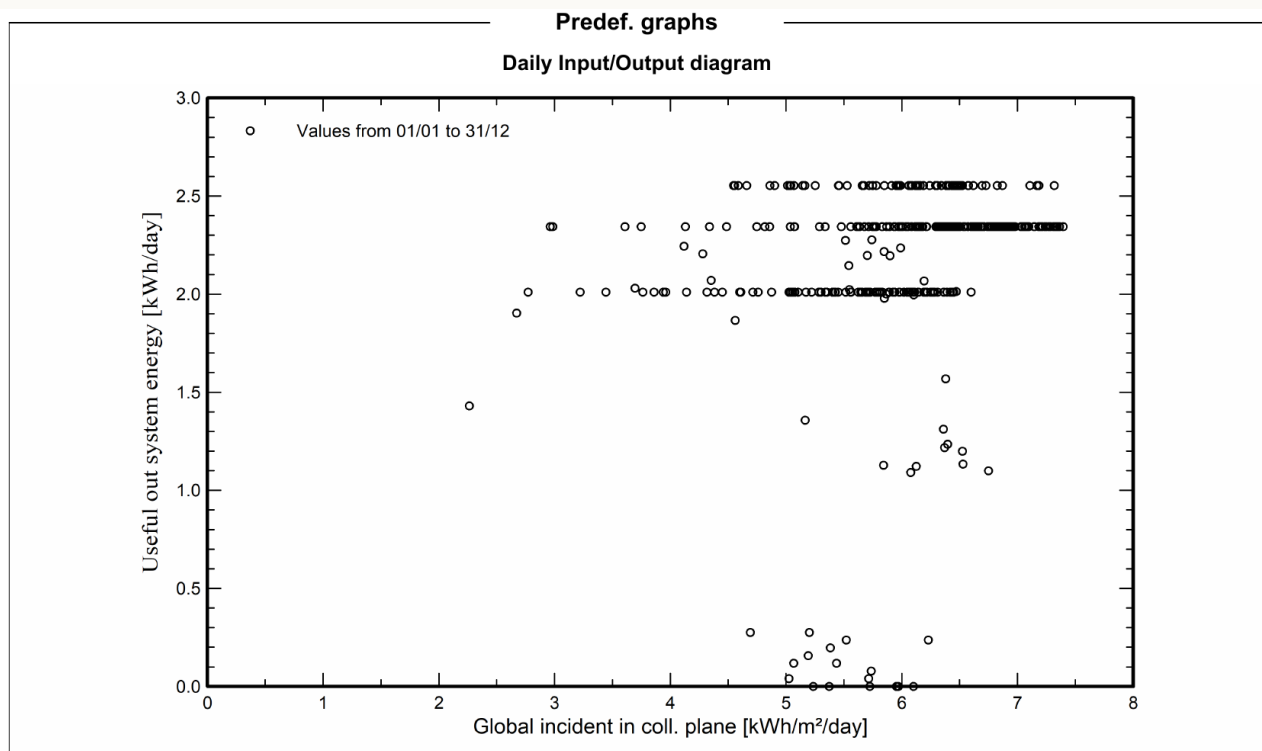
05 Energy Loss Diagram

The diagram below shows the energy flow path, starting from the incident solar irradiation through to the final energy delivered to the user, detailing each loss percentage along the way (thermal loss, wiring loss, converter loss, battery loss, etc.).



The two largest sources of loss in the system are: thermal loss in the PV modules (11.6%) due to high ambient temperatures at the site, and combined converter/battery losses during operation, charging, and discharging (~7%).

06 Daily Production Diagram & Discussion



Relationship between daily incident solar irradiation and useful energy delivered to the user throughout the year

07 Conclusion & Recommendations

Strengths

- High solar fraction of 93%
- Minimal excess/unused energy (only 11 kWh/year)
- Good projected battery lifetime (SOW 90-96%)

Challenges

- Energy shortfall during December and September (highest loss-of-load months)
- High thermal loss due to local climate conditions
- Overall Performance Ratio is moderate (74.9%)

Future Recommendations

- Increase battery capacity to reduce winter energy shortfalls
- Improve rear ventilation of panels to reduce thermal losses
- Study solar tracking to improve overall efficiency

This report was generated based on PVsyst V7.4.8 simulation results — Off-Grid Photovoltaic System Design Project | Zagazig University